

Whale interactions with Alaskan sablefish and Pacific halibut fisheries: Surveying fishermen perception, changing fishing practices and mitigation



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ABSTRACT

Whale depredation occurs when whales steal fish, damage fish or damage fishing gear. In Alaska, killer whales (*Orcinus orca*) and sperm whales (*Physeter macrocephalus*) primarily depredate on demersal sablefish (*Anoplopoma fimbria*) and Pacific halibut (*Hippoglossus stenolepis*) longline fisheries. Quantitative data on whale depredation in Alaska is limited due to low fishery observer coverage and minimal depredation evidence left on longline fishing gear. This study utilized semidirected interviews ($n=70$) and written questionnaires ($n=95$) with longline fishermen to examine: (1) perceptions and experiences of whale–fishery interactions in Alaska, (2) effects of depredation on fishing practices, and (3) potential depredation mitigation measures. Eighty-seven percent of fishermen surveyed agreed that whale depredation became worse between 1990 and 2010. Respondents reported changing their fishing practices in response to depredating whales in several ways, including: traveling up to 50 nautical miles and ceasing hauling operations up to 24 h until the whales left the fishing grounds. Respondents fishing in western Alaska, primarily encountering killer whales, were forced to wait longer and travel greater distances than fishermen operating in central and southeast Alaska, regions more affected by sperm whales. Deterrent research, gear modifications and real-time tracking of depredating whales were solutions favored by study participants. Survey respondent answers varied based on areas fished, quota owned, years involved in the fishery and vessel size. This study presents the first statewide evaluation of fishermen's perception and knowledge of whale interactions with the Alaskan longline fleet and is a critical step toward developing baseline data and feasible depredation mitigation strategies.

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1. Introduction

Increased fishing effort in Alaska throughout the 20th century has expanded anthropogenic influences on marine ecosystems and marine predators [1–3]. Interactions and competition with other cosmopolitan apex predators is a logical outcome of increased human presence in and reliance on marine environments. This interaction is exemplified by the issue of killer whale (*Orcinus orca*) and sperm whale (*Physeter macrocephalus*) depredation on longline fisheries in Alaska. Killer whale and sperm whale depredation occurs when whales remove fish from longline gear, damage fish or fishing gear. In Alaska, killer whales and sperm whales primarily depredate on high-dollar demersal longline fisheries such as sablefish (*Anoplopoma fimbria*) and Pacific halibut (*Hippoglossus stenolepis*). Killer whale interactions with longline

fisheries occur in four main regions in Alaska: the Bering Sea, Aleutian Islands, Western Gulf of Alaska and the coastal waters of Prince William Sound [4–6]. Sperm whale depredation predominates in the eastern Gulf of Alaska through Southeast Alaska [6,7] (Fig. 1).

Whale depredation has negative consequences for fishermen, management agencies and the whales themselves. Depredation can lead to significant economic losses for fishermen in the form of reduced catch, increased operating costs and damaged gear [8–11]. Overall catch rates can decline by as much as 30% and individual sets by 100% in the presence of depredating whales [6,12,13]. The potential effects that depredation may have on the accuracy of fish stock abundance indices is a critical issue for management agencies with important implications for fishermen [9,14,15]. Depredating whales may get a relatively easy meal of fish hooked on longline gear, but this activity can also have harmful consequences for the whales [16,17]. Whales may be at greater risk of mortality or injury from vessel strikes, risk of entanglement in gear and fishermen's frustration [8,17–19]. For instance, in Prince

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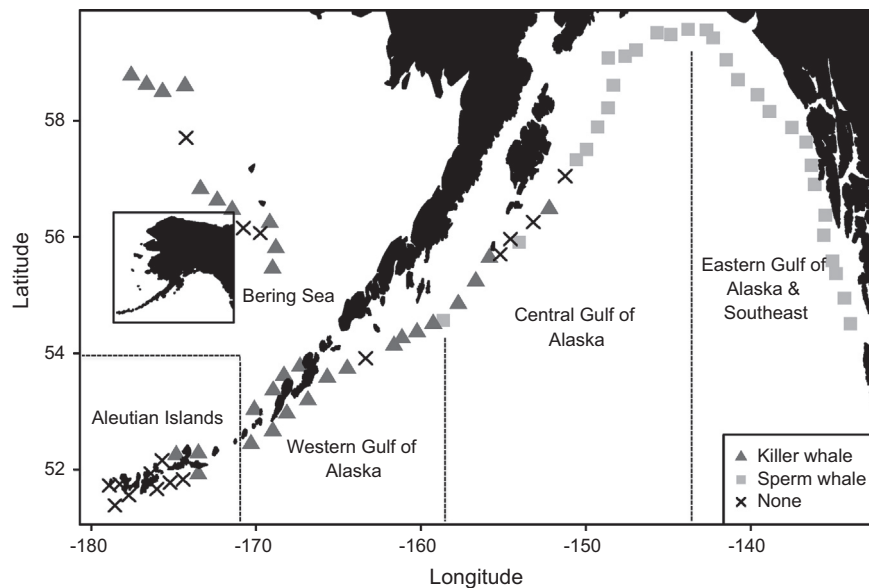


Fig. 1. Whale depredation by species and management areas based on the National Marine Fisheries Service (NMFS) longline survey, 1998–2011. NMFS longline survey locations mirror true longline fishery grounds along the continental slope.

William Sound in the 1980s, killer whale groups known for their interactions with longline vessels were photographed with bullet holes in their dorsal fins [20]. There is also a risk associated with modifying marine mammal foraging behavior and energy balance toward an unnaturally available and unreliable prey resource [19].

1.1. Toothed whales in Alaska

Killer whales and sperm whales are the two largest species of odontocetes or toothed whales. They are also the most cosmopolitan marine mammals with ranges spanning the globe [21,22]. Killer whales and sperm whales are K-selected species with large body size, long life spans, low natality rates and high maternal investment [23]. Killer whale depredation was documented as early as the 1960s by Japanese longliners in the Bering Sea [4]. It was estimated in 2010 that a minimum of 1300 resident killer whales inhabit the Bering Sea, Aleutian Islands and Western Gulf of Alaska [22]. However, more recent photographic mark-recapture assessments indicate that significantly more (perhaps twice this number) fish-eating killer whales use the coastal waters around the eastern and central Aleutians alone in some years [24]. Stock structure and abundance data are relatively deficient for north Pacific sperm whales [21], and as of 2012, they were listed as “endangered” under the 1973 Endangered Species Act and “depleted” under the 1972 Marine Mammal Protection Act. Depredation by sperm whales in Alaska is presumed to be primarily by solitary mature males from the North Pacific stock [25]. Whereas killer whale depredation has a longstanding history in Alaska [4], sperm whale depredation in the Gulf of Alaska is a relatively more recent phenomenon. Documented interactions with sperm whales may have increased in the late 1990s following the implementation of individual fishing quotas (IFQ) and associated extended sablefish and Pacific halibut fishing seasons [26].

1.2. Fishery background

The problem of toothed whale depredation is particularly acute in Alaska, where lucrative longline fisheries are prosecuted in areas supporting some of the world's greatest densities of fish-eating killer whales in western Alaska and an anecdotally increasing population of sperm whales in the Gulf of Alaska [24,26]. The

Pacific halibut fishery is managed jointly by the US and Canada under the International Pacific Halibut Commission. The Alaska sablefish fishery is managed by the North Pacific Fisheries Management Council (NPFMC). During the mid-1970s both fisheries experienced a significant reduction in season length, resulting in “derby-style” fishing with openers as short as 24 h [27,28]. Due to shorter fishing seasons and increased harvesting capabilities of larger vessels, the NPFMC implemented an IFQ program in federal waters off Alaska for sablefish and Pacific halibut in 1995 [27,29,30]. These fisheries have experienced significant consolidation since the inception of the IFQ program, and the number of vessels participating in the fisheries declined dramatically from 1995 to 2011, by about 70% for both sablefish and halibut [31]. The sablefish and Pacific halibut commercial longline fisheries are typically open for 9 months, from mid-March to mid-November. The majority of sablefish quota is harvested in April and May, whereas most halibut quota is harvested in June [31]. In 2011, standard ex-vessel prices averaged \$5.25 per pound for sablefish and \$6.60 per pound for Pacific halibut [32].

Pacific halibut fisheries are almost entirely prosecuted using fixed longline gear as Pacific halibut catch rates in pot fishing gear are relatively low [33]. Sablefish can effectively be fished with pot gear; however, pot fishing for sablefish or Pacific halibut is currently only allowed in the Bering Sea and Aleutian Islands management areas [34]. The sablefish fishery in the Bering Sea has seen a large number of vessels transitioning to pot fishing due in part to killer whale depredation [34]. In 2000, the pot fishery accounted for less than 10% of the fixed gear sablefish catch in the Bering Sea and Aleutian Islands, whereas in 2009 pot fishing accounted for over 70% of sablefish catch in the Bering Sea [34]. Sablefish and Pacific halibut fisheries in the Gulf of Alaska management areas are prosecuted with fixed longline gear only [35]. In 2012, the total allowable catch (TAC) for Pacific halibut was 23.3 million net pounds (headed and gutted) and 26.5 million round pounds for sablefish, with the highest catches occurring in the Central Gulf followed by the Western Gulf and Southeast Alaska [36]. The percentage of quota landed is generally close to 100% in most management areas, with the exception of Bering Sea sablefish and Pacific halibut fisheries, which typically have not been harvested to the full extent of the total allowable catch in recent years [31,34,37].

This study of fishermen's knowledge uses a mixed methods approach of semi-directed interviews and written questionnaires to examine how fishermen perceive whale interactions, the way in which depredation shapes fishing behavior and potential depredation mitigation solutions. Fishing knowledge is based upon experience and interactions with marine environments through long-term observation and direct interaction with marine resources [38–40]. Studies incorporating fishermen's local knowledge are gaining traction in conventional scientific and policy forums [39,41,42]. Fishermen's knowledge can elucidate long-term time trends and spatial components to systems not captured within traditional data sets [43–45]. Fishermen's knowledge can also be useful in studies examining topics, such as whale depredation, where data is deficient [42]. Previous studies of depredation in Alaska are limited and have focused on quantifying the magnitude of depredation, relying primarily on biological survey data [6,7,46]. To our knowledge, this study serves as the first detailed examination of whale depredation in Alaska using social science research methods, focusing on fishermen's perception and experience based on results from interviews and written surveys collected across Alaska from 2010 to 2012.

2. Methods

2.1. Interviews

Semi-directed interviews ($n=70$) were conducted with longline fishermen from July 2010 through December 2011. Semi-directed interviews enable a researcher to guide an interview, but provide space for participants to dictate the direction and scope of the interview [47,48]. Informants were selected using snowball sampling methods. Snowball sampling is a nonprobability method for selecting respondents, often based on their representativeness or fishing experience [49]. In this study, colleagues and fisheries managers recommended fishermen with long-term experience in the longline fishery. These fishermen in turn recommended other fishermen with long-term experience in the Alaskan longline fleet. Interviews ranged in length from 30 min to 150 min. Interviews were conducted in-person and recorded via livescribe with permission when logistically feasible. Remaining interviews were conducted over the phone and were recorded with permission via skype call recording. Recorded interviews ($n=55$) were then transcribed and uploaded into Atlas.ti, a qualitative data analysis software. Interview data were list-coded for thematic content based on inductive coding methods and a grounded theory approach [47,50,51]. These techniques are used to identify concepts and themes that appear in the interview text and link emergent concepts together to develop formal theories [50]. Interviews were analyzed for code frequency and code co-occurrence.

2.2. Written survey instruments

Results from a literature review and thematically coded interviews were used to construct a written survey instrument distributed to longline fishermen in 2011 and 2012. A purposive sample of respondents ($n=95$) was selected based on long term fishing experience and, therefore, was deliberately non-random [39,49,52]. This nonprobability method limits the theoretical basis for drawing fleet-wide inferences and for measuring variability and bias [49]. Despite limitations associated with nonprobability sampling, selectively targeting respondents with long-term fishing engagement enables researchers to work with fishing participants with the most experience in and “direct links” to the marine environment [39]. The written survey was distributed in-person to experienced Alaskan quota holders across a wide geographic range

of fishing communities in Alaska and Washington at community meetings, longline fishing association and regional fishery management meetings. Survey distribution generally followed an introductory presentation about the subject matter and the research project. The distribution of reported survey respondent effort by management area and quota harvested by vessel category closely mirrored the 2012 true fishery breakdown for quota units harvested by management area and vessel categories [53], indicating a sample that was representative of basic true fishery characteristics.

The written survey was divided into four sections: demographic and fishing information, perceptions of the depredation problem, strategies for depredation avoidance and suggested solutions. In the survey, specific management areas were grouped into three main regions: (1) Western Alaska (Bering Sea, Aleutian Islands and Western Gulf) (2) Central Gulf, and (3) Southeast Alaska (Fig. 1). Written survey question methods included Likert scales, true–false, multiple-choice and open-ended questions [54–57]. Survey questions were analyzed for basic trends and to explore hypotheses addressing response differences by management area, region, species most often encountered (sperm whales or killer whales) or vessel size.

A generalized linear proportional odds model for ordered categorical data was used to investigate factors shaping fishermen's depredation avoidance practices. Proportional odds models require fewer parameters and are commonly used to estimate the odds ratio of the cumulative probabilities of responses for ordered categorical data [49,58,59]. Four models were run to address each question individually: (1) average time fishermen waited to avoid depredating whales per season, (2) average number of times per season fishermen were forced to wait, (3) distance fishermen traveled to avoid depredating whales, and (4) average number of times per season fishermen were forced to travel due to whales. Based on interview results, whale species most often encountered and vessel size were hypothesized to influence changed fishing practices. Additional explanatory variables considered included primary fishing region, total quota owned and years fishing. The final model for each question was selected based on lowest Akaike Information Criterion (AIC) score and model diagnostics [58–60].

Individual Likert-type items with the same directionality were grouped together by theme and summed to create composite Likert scales [54–56] with four focus areas: depredation trends (assessing how fishermen perceived temporal trends); personal impacts (measuring the severity of impacts that fishermen experienced due to whale interactions); solutions (evaluating how fishermen perceive depredation solution options); changing fishing practices (investigating whether fishermen significantly altered their fishing behavior in response to depredating whales). These four Likert scale composites were analyzed using linear regression methods [49,60]. Final models were selected based on stepwise regression and model diagnostics [60].

3. Results

3.1. Interviews

Fifty-five interviews were transcribed and analyzed in Atlas.ti. In total, 921 interview excerpts were assigned a specific code. Four themes were identified during qualitative data analysis: changing fishing practices, management issues (e.g. stock assessment, quota allocation), depredation trends and whale behavior (Fig. 2). Changing fishing practices was broken down into three sub-categories: shifted fishing seasons, depredation avoidance measures to evade whale interactions, and deterrent measures to confuse or disturb the whales. Example deterrents included seal bombs and acoustic harassment devices. Forty codes were designated under the four

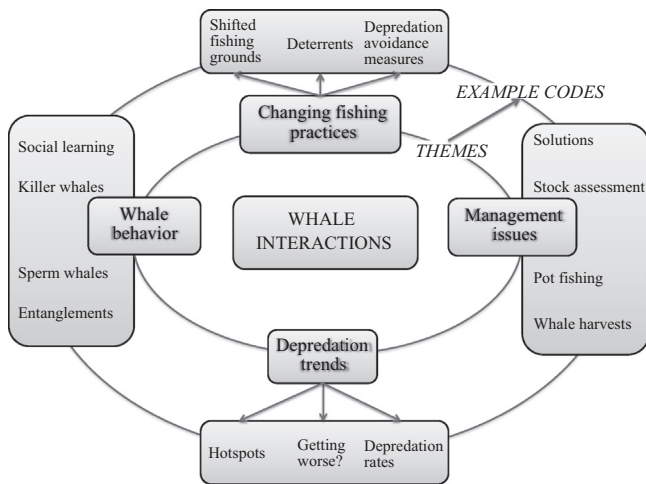


Fig. 2. Themes and example codes used during interview data analysis.

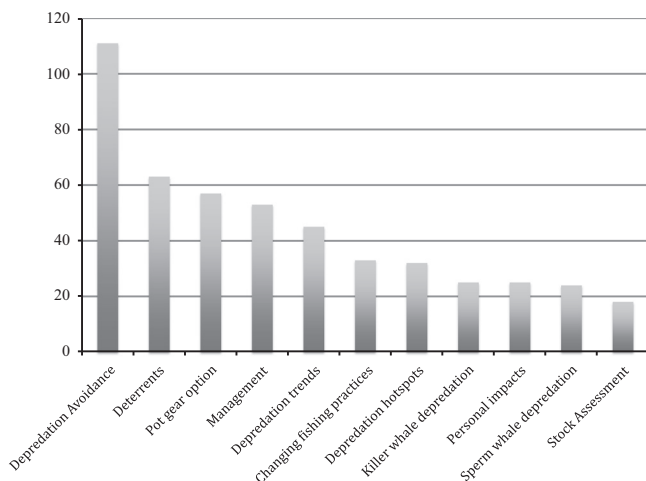


Fig. 3. Code occurrences in combined interview text from 55 transcribed interviews ($n=921$ codes assigned).

themes. “Depredation avoidance”, “deterrents”, “pot gear”, “management”, and “depredation trends” were the most commonly occurring codes across all interviews (Fig. 3).

The highest code co-occurrence rates were: “changing fishing practices” and “depredation avoidance ($r=0.17$)”, and “personal impacts” and “depredation impacts on quota ($r=0.15$)”. “Pot gear” co-occurred most commonly with “management ($r=0.10$)” and “depredation solutions ($r=0.11$)”.

3.2. Written surveys

3.2.1. Respondent demographics

A total of 95 longline fishermen completed written questionnaires. Combined, these fishermen fished approximately 6.4 million pounds of Pacific halibut quota (27% of overall halibut quota in 2012) and 5.8 million pounds of sablefish (20% of overall sablefish quota harvested in 2012). The average number of years respondents were involved in the longline fishery was 22.6 years for a sum total of 1990 years for all respondents. The majority (92%) of respondents were male, and the average age of all respondents was approximately 47.5 years. Survey respondents reported primary fishing effort in the Central Gulf (36%), Southeast Alaska (31%), Western Gulf (17%), Bering Sea (10%) and Aleutian Islands (6%). The total amount of quota owned by respondents was not correlated with the number of years spent fishing or age (Pearson

correlation test, $p > 0.50$) but was significantly different among management areas (ANOVA, $F=4.4$, $p=0.003$), with the median quota higher for fishermen operating primarily in the Western Gulf and Bering Sea. Only six of the 95 respondents reported having fished pot gear for sablefish at some point during their career. Respondents reported the majority of quota (lbs) was caught on catcher vessels less than 60 ft (category C; 44%), followed by catcher vessels greater than 60 ft (B; 39%), freezer vessels (A; 12%) and catcher vessels less than 35 ft (D; 4%), which corresponded closely with the true fishery breakdown of quota harvested by vessel category [36].

3.3. Fishermen experience and perception

3.3.1. Depredation trends

The majority of respondents strongly agreed (78.3%) or mostly agreed (8.7%) that whale depredation became worse between 1990 and 2010, and most respondents strongly agreed (57.6%) or mostly agreed (16.3%) that they were frustrated by whale depredation. Respondents (65.6%) cited “more whales” or “whales learning the behavior” as the primary reasons for depredation becoming more severe as an interaction. “Longer fishing seasons [associated with the IFQ program]” (11.8%) was also noted as a reason for the interaction becoming more problematic for longline fishermen in Alaska. Respondents were asked which species of whale generally depredated where they fished. Just over half of the respondents reported interactions primarily with sperm whales (56.7%). The remaining 41 respondents (43.1%) reported encountering mostly killer whales. The species fishermen described interacting with most frequently was significantly correlated with the area where they reported fishing most (Pearson's Chi-squared test, $\chi^2=47.5$, $p < 0.001$); with fishermen in the Bering Sea, Aleutian Islands and Western Gulf confronting killer whales and fishermen in Southeast interacting almost entirely with sperm whales.

Survey respondents were asked an open-ended question to list any marine environment characteristics they associated with whale depredation. The majority of respondents (55%) that responded to the question ($n=42$) listed “high catch rates,” suggesting whales target fishing grounds with higher catch per unit effort (CPUE). Additional factors reported to contribute to depredation included, “season” or “time of year” and “depth,” with depredation generally thought to be more severe in the spring in April or May and along the shelf edge at depths greater than 100 fathoms for sperm whales (Fig. 4). Fishermen expressed “loss of

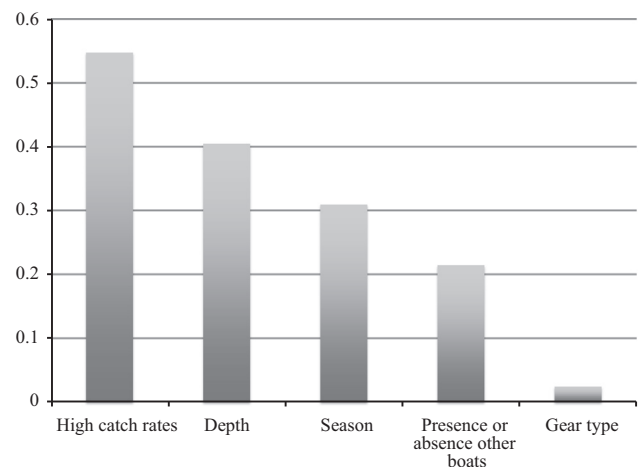


Fig. 4. Commonly listed factors influencing the number of whale interactions. Proportions are based on the set of respondents that listed at least one factor.

catch” associated with whale depredation was their biggest concern (34.7%), followed by “inaccurate stock assessments (17.9%).” Eighty-three percent of respondents strongly agreed or mostly agreed that whale depredation is reducing the accuracy of sablefish and Pacific halibut stock assessments.

3.3.2. Depredation rates: sperm whales versus killer whales

Survey respondents mostly encountering sperm whales reported seeing on average one to five sperm whales around the vessel when depredation occurred (93.6%). Respondents interacting mostly with killer whales reported an average of six to 10 killer whales around the vessel during depredation events (52.6%). The majority of respondents (31.2%) estimated the overall percentage of sets depredated was 10–25%. There was no significant difference in the reported proportion of sets depredated between whale species or among fishing regions (Fisher's exact test, $p > 0.05$). There were, however, significant differences in the reported depredation rate (catch removal) on individual sets when whales were present, based on whale species and region (Fisher's exact test, $p < 0.001$). The majority of respondents (70.7%) that reported interactions with killer whales (primarily western Alaska) estimated that depredation rates exceeded 40% of catch. In contrast, the majority of respondents (42.6%) experiencing interactions with sperm whales (eastern Gulf of Alaska) reported that depredation rates on individual sets were less than 20% of catch. Sablefish and Pacific halibut were the species most targeted by whales according to survey respondents. Additional groundfish species whales targeted included arrowtooth flounder (*Atheresthes stomias*), Greenland turbot (*Reinhardtius hippoglossoides*), giant grenadier (*Albatrossia pectoralis*), and Pacific cod (*Gadus macrocephalus*).

3.4. Changing fishing practices

3.4.1. Depredation avoidance

Survey respondents confirmed that fishermen are forced to change the way they fish in response to the threat or presence of depredating killer whales and sperm whales. Most respondents (64.1%) strongly agreed or agreed that they often had to fish less efficiently to avoid depredating whales, and 88.0% agreed or strongly agreed that they were constantly on the lookout for whales when fishing. Respondents reported that “moving to different fishing grounds (36.3%)” and “dropping the gear to wait the whales out (37.4%)” were their preferred methods to avoid depredating killer whales and sperm whales. A small percentage of respondents did report they preferred to fish through the set when whales were present (12.1%). Respondent answers varied by fishing region, and fishermen operating in central and eastern Gulf were more likely to

report “moving to a different site” (Fisher's exact test, $p < 0.001$). Other avoidance measures selected included “fishing in tandem with other boats” (34.2%) and “hauling the gear quickly” (30.1%).

Respondents were asked to estimate the frequency and extent of depredation avoidance measures they employed throughout the season; including the amount of time and how often they were forced to extend their gear soak time if whales showed up on the gear during haul-back operations and how far and how often they traveled to avoid depredating whales (Table 1). The majority of respondents fishing in central and eastern Alaska reported wait times less than 12 h (68.3%) and forced travel distances generally less than 25 nautical miles (63%) in the presence of depredating whales (Table 1). Respondents fishing in western Alaska (Bering Sea, Aleutian Islands and Western Gulf) generally reported wait times greater than 12 h (50.0%) and travel distances greater than 25 nautical miles (69.0%; Table 1).

3.4.2. Deterrents

Above we defined depredation avoidance as evasion methods employed once whales have already begun to depredate on set fishing gear. Deterrents were defined as aversive stimuli introduced to the whales' environment to deter or confuse the whales so that depredation may be less likely to occur. One interview respondent noted that deterrents are often ineffective, as whales learn to adapt:

[We've tried] all kinds of electronic sounds, music, whatever; anything that would confuse them. But whatever you do, it seems they adapt to it. They know the sound of everybody's propeller screw out there.

Interview respondents reported using acoustic devices such as echolocation blockers or killer whale sound playbacks, targeted sonar, seal bombs and dummy sets as deterrents. Most survey respondents answered that “acoustic deterrents (50.0%)” had the best chance of being an effective deterrent, followed by gear modifications (29.0%).

3.5. Management and solutions

The issue of whale interactions with sablefish and Pacific halibut fisheries has implications for fishermen and fishery managers. There is concern amongst both groups that whale depredation negatively impacts the ability of fishery managers to accurately assess fish stocks. The majority of respondents strongly agreed (58.7%) or agreed (25.0%) that whale depredation reduces the accuracy of fish stock assessments. Over half of all respondents (56.5%) and 74.1% of western Alaska respondents disagreed or strongly disagreed with the statement that “managers have been up front and proactive in dealing with whale depredation”.

Table 1

Count of estimated wait durations, average number of wait events per season, estimated distances traveled, and average number of times distance traveled per season broken down by region primarily fished. Fisher's exact test results for each question and region.

	Estimated wait duration				Average number of waiting events per season			
	0–3 h	4–12 h	13–24 h	> 24 h	1–5 times	5–10 times	11–15 times	> 15 times
West	3	10	7	9	6	11	8	2
Central	10	18	1	3	22	8	1	0
Southeast	8	7	1	1	14	1	0	0
Total	21	35	9	13	41	20	9	2
	Fisher's exact test, $p=0.004$				Fisher's exact test, $p < 0.001$			

	Estimated distance traveled to avoid whales			Average number times distance traveled per season			
	< 25 nm	25–50 nm	> 50 nm	1–5 times	5–10 times	11–15 times	> 15 times
West	9	12	8	7	12	7	3
Central	18	12	0	22	7	0	0
Southeast	13	5	1	16	2	0	0
Total	38	29	9	45	21	7	3
	Fisher's exact test, $p < 0.001$			Fisher's exact test, $p < 0.001$			

When respondents were asked about management solutions to reduce the effects of whale depredation, 83% answered that “real-time tracking of whales” was a preferred management option, and 76.1% of respondents agreed or strongly agreed that they would be “interested in real-time tracking of groups of depredating whales by fishermen”. Respondents had the option to list additional management measures not included on the survey question. Of those participants that responded to this optional open question ($n=21$), 33.3% listed “pot fishing” as their preferred management solution followed by “acoustic deterrents” (19.0%). Respondents were also asked the open question, “What could managers do to assist fishermen dealing with whale depredation?” The majority of participants who listed management solutions ($n=54$) focused on additional deterrent research (29.6%), the flexibility to switch to pot gear (18.5%) and additional whale abundance and movement research (16.7%). Real time tracking of whales, limited whale harvests and federal reimbursement programs for losses were also commonly listed by respondents as potential management solutions (Fig. 5).

The option to transition to pot gear is relevant to discussions of depredation avoidance, depredation mitigation and management. Interview and survey respondent opinions were divided on the issue of opening up sablefish to pot fishing in federal Gulf of Alaska waters. Respondents fishing in the Bering Sea, Aleutian Islands, and Western Gulf were more likely to consider the transition to pots a viable option. For instance, 32.3% of western Alaska respondents strongly agreed that the transition to pots was a possibility versus 11.1% of Southeast respondents. Although respondent answers to the feasibility of a pot transition for sablefish harvesting was not statistically different based on fishing region (Fisher's exact test, $p=0.666$), there was a significant relationship between vessel category and pot transition perception (Pearson's Chi-squared test, $\chi^2=21.1$, $p=0.049$). Fishermen operating B class (catcher > 60 ft) and C class (catcher 35–59 ft) vessels were more likely to agree that the transition to pot fishing was a possibility on their vessels.

3.6. Predicting perceptions and changing fishing practices

3.6.1. Likert scales

Individual Likert questions were grouped together to create composite Likert scales for further analysis. The impacts scale dealt with the severity of impacts that fishermen experienced due to whale interactions. Higher impact scale values represented strong

support or agreement that depredation significantly impacted continued fishing behavior and profitability of an individual. The AIC-best model for the impacts scale included fishing region as a categorical variable and years fishing as a continuous variable (ANCOVA, $R^2=0.17$, $p=0.002$).

$$\text{Perceived impacts scale} = \beta_0 + \text{area fished}_i + \beta_1(\text{years fishing}) + \epsilon$$

Respondent Likert scale responses decreased linearly with years fished (Fig. 6b) and were highest in western Alaska, followed by the central and southeast Alaska (Fig. 6a). The depredation impacts scale was significantly correlated with the changing fishing practices scale (Pearson's correlation test, $r=0.573$, $p < 0.001$), and depredation solutions scale (Pearson's correlation test, $r=0.456$, $p < 0.001$).

Area fished and years fishing also significantly impacted respondent answers to the Likert scale addressing changing fishing practices. Higher composite scores on the fishing practice scale indicated strong agreement that fishermen would alter the way they fish in response to depredating whales. The final linear model for the changing fishing practices scale included fishing region as a categorical variable and years fishing as a continuous variable (ANCOVA, $R^2=0.13$, $p=0.012$).

$$\text{Changing fishing practices scale} = \beta_0 + \text{area fished}_i + \beta_1(\text{years fishing}) + \epsilon$$

Respondents from western and central Alaska were more likely to change their fishing practices due to depredating whales (Fig. 6c), and the amount of years fishing also impacted responses to the changing fishing practices scale (Fig. 6d).

Respondent perception of depredation trends involved Likert questions assessing whether whale depredation is getting worse over time. The solutions scale evaluated how fishermen perceived various potential depredation mitigation solutions. The final linear model for the depredation trends scale (ANOVA, $R^2=0.09$, $p=0.004$) and depredation solutions scale models (ANOVA, $R^2=0.13$, $p < 0.001$), selected using stepwise regression methods, included total quota owned (halibut and sablefish quota summed) as a continuous variable.

$$\text{Perceived depredation trends} = \beta_0 + \beta_1(\text{total quota owned}) + \epsilon$$

$$\text{Perceived depredation solutions} = \beta_0 + \beta_1(\text{total quota owned}) + \epsilon$$

The more quota an individual owned the more likely they were to strongly agree that depredation is spreading and getting worse (Fig. 6e) and to express strong support of various management options including real-time tracking and deterrent use (Fig. 6f).

3.6.2. Predicting depredation avoidance and changing fishing practices

A proportional odds model for ordered categorical data was used to examine the factors influencing how respondents changed their fishing practices to avoid depredating whales, specifically how far and how often they would travel to avoid the whales and how long and how many times they would “wait the whales out” [49,58]. The best model for wait times, wait frequency and travel frequency questions included region fished (West, Central, East) and years fishing as explanatory variables.

$$\text{Logit (Wait length, Number of waits, or Number of motors)} = \beta_0 + \text{region}_i + \beta_1(\text{years fishing}) + \epsilon$$

The effect estimates from all three models suggested that the cumulative probability of respondents reporting a higher number of wait events or motor events increased with respondents fishing primarily in western Alaska, followed by central Alaska and Southeast respondents (Fig. 7a, b and d). A higher number of years fished was also associated with increases in the reported wait and motor events. The best model for the estimated average distance motored when whales were present included vessel

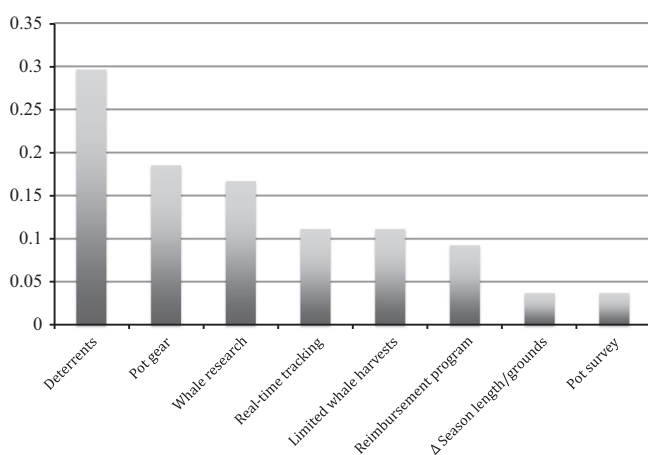


Fig. 5. Commonly listed management solutions to reduce the impact of whale depredation. Proportions are based on the set of respondents that listed at least one suggestion.

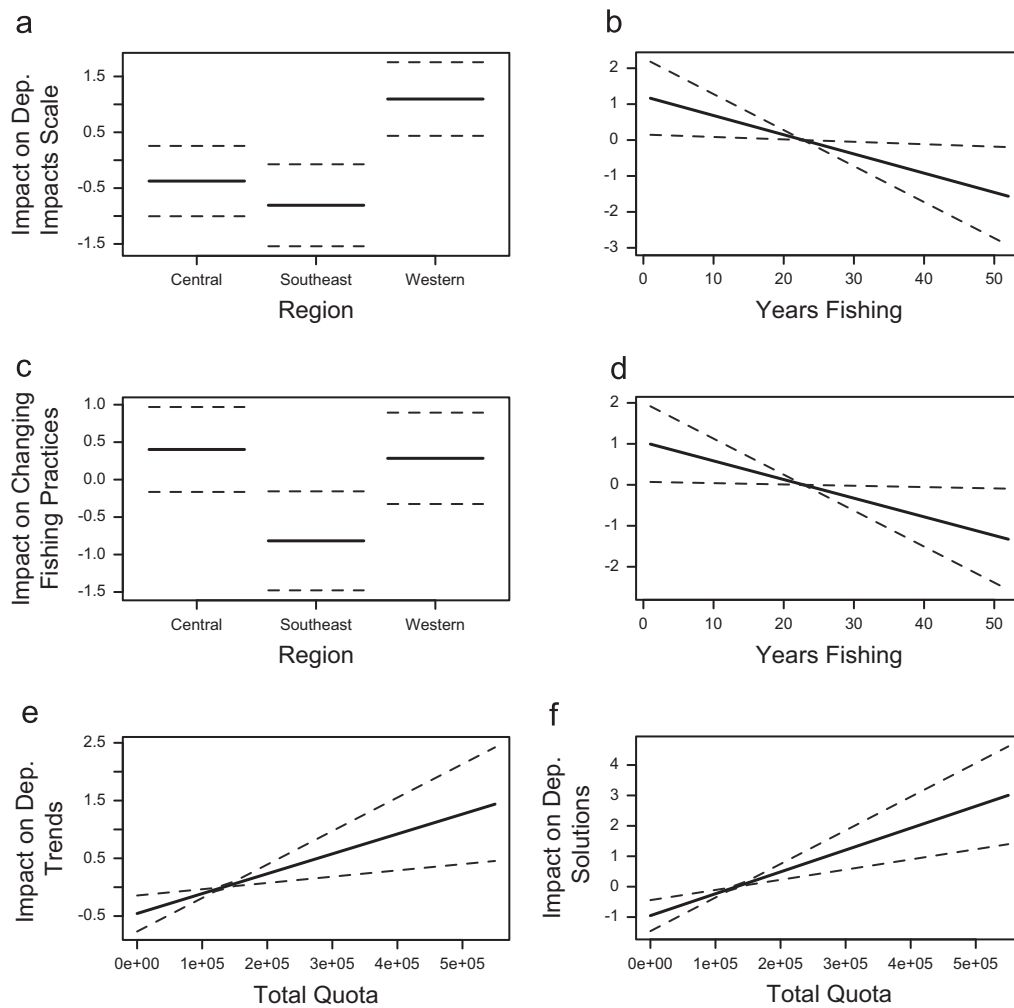


Fig. 6. Standardized effects of region and years fishing on Likert scales “Depredation Impacts (a, b)” and “Changing Fishing Practices (c, d)”. Effect of total quota on likert scales “Depredation Trends (e)” and “Depredation Solutions (f)”. Dotted lines denote standard errors.

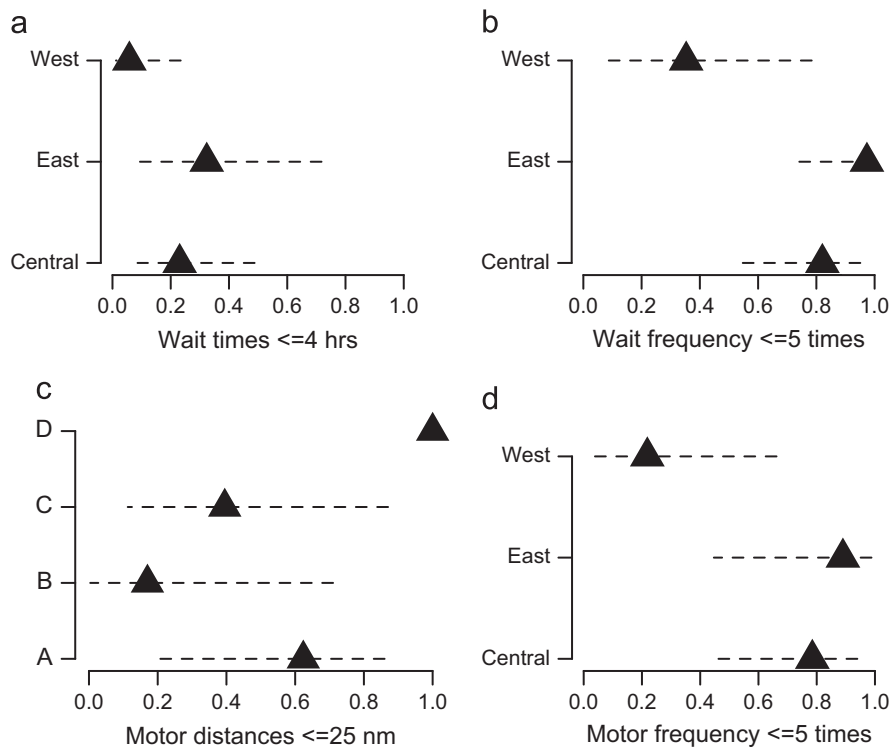


Fig. 7. Model-estimated cumulative probabilities of wait times ≤ 4 h (a), number of waiting events per season ≤ 5 times (b), motor distances ≤ 25 nautical miles (c) and ≤ 5 motor events per season (d) broken down by region or vessel category ($n=95$) with 95% confidence intervals.

category, species most encountered (sperm whales versus killer whales) and years fishing as explanatory variables.

Logit (Average distance traveled) = $\beta_0 + \text{whale species}_i$
 + vessel category_j + $\beta_1(\text{years fishing}) + \varepsilon$

For this model, the cumulative probability of a respondent traveling longer distances was significantly higher for respondents dealing primarily with killer whales and respondents fishing larger B class vessels (> 60 ft). Respondents fishing smaller D class vessels (< 35 ft) were more likely to report shorter distances traveled (Fig. 7c).

4. Discussion

Depredation by killer whales and sperm whales on longline Pacific halibut and sablefish fisheries is a relatively common occurrence in Alaskan waters and is changing the way the fisheries are pursued. Survey respondents estimated that on average 10%–25% of sets are impacted by whale interactions; however, the direct catch losses associated with whale depredation varied significantly by species and region fished. Reported depredation rates and catch removals associated with killer whales (> 40%) tended to be higher than sperm whale catch removals (< 20%). The majority of interview and written questionnaire respondents expressed that depredation by both killer whales and sperm whales became more severe in the last 20 years due to an increasing number of whales learning the depredation behavior. Many respondents also noted that with the implementation of the IFQ program and extension of the fishing season to 9 months there are more opportunities for whales to learn depredation behavior throughout the longer season. Questionnaire respondents reported that whales targeted fishing grounds with higher catch rates. This link between high catch rates and whale depredation has been shown in previous studies investigating sperm whale depredation on longline Patagonian toothfish (*Dissostichus eleginoides*) fisheries in the Southern Ocean [15,61–63] and more recently in a study that linked killer whale depredation to areas of high sablefish catch rates in western Alaska [6].

Fishermen primarily operating in Southeast (generally encountering sperm whales) perceived whale depredation impacts to be less severe than fishermen operating in western and central Alaska (Fig. 6a) and were the least likely to agree that they changed their fishing practices due to depredating whales (Figs. 6c, 7). Fishermen with more years of fishing experience stated that depredation had relatively lower impacts on their desire to remain in the fishery (Fig. 6b), and they were less likely to report changing their fishing practices (Fig. 6d). This is consistent with interview findings that respondents with longstanding experience in the fishery perceived that changes to fishing practices were ineffective at avoiding whales. It is also not surprising that respondents with a demonstrated long-term history in the longline fishery would be less likely to consider leaving the fishery (due to whale interactions or other issues). Owners of higher quota amounts agreed more strongly with the statement that depredation is spreading and getting worse (Fig. 6e) and that depredation mitigation solutions such as real-time tracking of depredating killer whales and deterrents could reduce the severity of whale interactions with the longline fleet (Fig. 6f). Thus, the personal financial investment (quota owned) of study participants may be linked to perceptions of depredation severity and the importance of deterrent research.

This study confirms that fishermen are forced to change their fishing practices to avoid depredating whales, resulting in reduced overall economic efficiency for the Alaskan longline fleet. The financial impacts of whale depredation have been estimated in other fisheries where interactions occur. Between 2003 and 2008, killer whales and sperm whales were responsible for \$6.1 million

losses in the Crozet Islands Exclusive Economic Zone (EEZ) and an annual loss of \$2.1 million in the Kerguelen Islands EEZ based on the current market price of Patagonian toothfish [19,63]. Despite the economic challenges posed by whale depredation, respondent answers to questions about the impact of depredation on their desire to remain in the fishery were mixed. Approximately 40% of respondents agreed that depredation could affect their decision to continue fishing quota, 28% were neutral and 31% disagreed that depredation would influence their decision to continue fishing.

This study also investigated fishermen perception of depredation management and mitigation solutions. Over 80% of respondents expressed concern that whale depredation was reducing the accuracy of the sablefish and Pacific halibut stock assessments. As one interview respondent from the Bering Sea stated:

This is our living. We don't want to knock the stocks down. We don't want the whales to knock the stocks down. We want healthy, sustainable stocks that we can take our part of but still have a sustainable biomass. It's critically important for us to know what's out there, both on the survey and knowing how much the whales are actually taking.

The majority of respondents also expressed that management agencies had not been proactive and up-front in dealing with depredation, indicating a degree of frustration amongst respondents with regards to the current depredation management strategies employed by the IPHC and NMFS. A Central Gulf longliner noted during an interview: "It's [whale depredation is] so touchy that it's almost like the elephant in the room."

Respondents listed deterrent research, allowing the switch to pot fishing in the Gulf of Alaska, whale abundance/distribution studies and real-time tracking of depredating whales as management or research measures they would like to see agencies place more emphasis on going forward. In interviews and open-ended questions, respondents expressed an interest in cooperative research and improving the lines of communication with management agencies. There are inherent challenges that fishermen and government agencies would face in cooperatively addressing a topic involving charismatic megafauna (whales) and endangered marine mammals and lucrative longline fisheries with long-term investments by fishery participants. A Central Gulf longliner touched on this issue during an interview:

If the fishermen and the scientists work together... and I've seen it happen, and it's such a great thing to be involved in. If we could work together, but this whole whale thing is such an emotional issue. And whales are more valuable than people.

Pot gear is currently allowed only in the Bering Sea and Aleutian Islands, and killer whale depredation has been a factor in a number of vessels changing from longline gear to pot gear in the Bering Sea [34]. The transition to pot fishing gear was a commonly noted depredation mitigation option by survey and interview respondents. Pot gear is currently not susceptible to killer whale or sperm whale depredation; however, interview and questionnaire results demonstrated that opening up the Gulf of Alaska to pot fishing for sablefish was a contentious topic for the longline fleet. For instance, pot gear can be used to harvest sablefish but is not suitable for effectively harvesting Pacific halibut at this time. There is a risk that the Pacific halibut fishery would, therefore, incur increased depredation pressure if pot gear were legalized for sablefish in the Gulf of Alaska. Gear conflict was an additional concern associated with allowing pot gear in the Gulf of Alaska. A longliner from the Western Gulf noted:

The two fisheries don't mix. They wouldn't overlap. Because if you try to fish black cod pots where another boat is fishing longline gear, the longline gear always loses. And unless you go to halibut pots too, you are just going to be pushing the sperm whales on to another fishery.

Despite the complexities discussed above, many study participants supported some degree of pot fishing for sablefish in the

Gulf of Alaska. The NPFMC is currently conducting reviews to determine the feasibility of reintroducing pot fishing for target groundfish species in federal Gulf of Alaska waters.

5. Conclusions

This study represents the first statewide survey of Pacific halibut and sablefish fishermen interactions with killer whales and sperm whales. The sample for this study was selected non-randomly, based on fishing experience, and therefore fleet-wide conclusions cannot be drawn based upon these results. Despite these limitations, respondents accounted for over 19% of sablefish and 26% of halibut quota and were representative of basic true fishery characteristics including reported effort by management area and quota harvested by vessel category [53]. Results suggest that the impacts of depredation are widespread in Alaska, but there are significant regional differences in killer whale and sperm whale depredation. The severity of killer whale depredation in western Alaska necessitates more extreme depredation avoidance measures and altered fishing practices. Additionally, the ability of individual vessels to avoid depredating whales and change fishing practices differed based on vessel size, gear type and the amount of quota harvested. Fishery policies, such as the transition to sablefish and Pacific halibut IFQs in 1995, can have unforeseen long-term consequences and may be linked to the spread of the whale depredation behavior in Alaska. Future management actions should recognize regional and species-level differences in whale depredation and vessel adaptability and incorporate studies estimating costs to the fleet and depredation mitigation options.

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References

- [1] Mansfield B. Modern industrial fisheries and the crisis of overfishing. In: Peet R, Robbin P, Watts M, editors. *Global political ecology*. Abingdon, Oxon: Routledge; 2011. p. 84–99.
- [2] Pauly D, Watson R, Alder J. Global trends in world fisheries: impacts on marine ecosystems and food security. *Philosophical Transactions of the Royal Society B: Biological Sciences* 2005;360(1453):5–12.
- [3] Read A. The looming crisis: interactions between marine mammals and fisheries. *Journal of Mammalogy* 2008;89(3):541–8.
- [4] Dalheim M. Killer whale (*Orcinus orca*) depredation on longline catches of sablefish (*Anoplopoma fimbria*) in Alaskan waters. Northwest and Alaska Fisheries Center Processed Report 88-14.
- [5] Matkin C. Killer whale interactions with the sablefish longline fishery in Prince William Sound, Alaska 1985 with comments on the Bering Sea. Juneau, AK: National Marine Fisheries Service; 1986. p. 1–12.
- [6] Peterson M, et al. Killer whale (*Orcinus orca*) depredation effects on catch rates of six groundfish species: Implications for commercial longline fisheries in Alaska. *ICES Journal of Marine Science*; in press.
- [7] Sigler MF, et al. Sperm whale depredation of sablefish longline gear in the northeast Pacific Ocean. *Marine Mammal Science* 2008;24(1):16–27.
- [8] Ashford JR, Rubilar PS, Martin AR. Interactions between cetaceans and longline fishery operations around South Georgia. *Marine Mammal Science* 1996;12(3):452–7.
- [9] Purves MG, et al. Killer whale (*Orcinus orca*) and sperm whale (*Physeter macrocephalus*) interactions with longline vessels in the patagonian toothfish fishery at South Georgia, South Atlantic. *CCAMLR Science* 2004;11:111–26.
- [10] Yano K, Dalheim M. Behavior of Killer Whales *Orcinus Orca* during longline fishery interactions in the Southeastern Bering Sea and adjacent waters. *Fisheries Science* 1995;61(4):584–9.
- [11] Goetz S, et al. Experimental fishing with an “umbrella-and-stones” system to reduce interactions of sperm whales (*Physeter macrocephalus*) and seabirds with bottom-set longlines for Patagonian toothfish (*Dissostichus eleginoides*) in the Southwest Atlantic. *ICES Journal of Marine Science: Journal du Conseil* 2011;68(1):228–38.
- [12] Dalla Rosa L, Secchi E. Killer whale (*Orcinus orca*) interactions with the tuna and swordfish longline fishery off southern and south-eastern Brazil: a comparison with shark interactions. *Journal of Marine Biological Association UK* 2007;87:135–40.
- [13] Sivasubramaniam K. Predation of Tuna longline catches in the Indian Ocean, by killer-whales and sharks. *Bulletin Fisheries Research Station, Ceylon* 1964;17(2):221–36.
- [14] Donoghue M. Report workshop on the depredation in the Tuna longline fisheries in the Indian Ocean, in workshop on the depredation in the Tuna longline fisheries in the Indian Ocean, Indian Ocean Tuna Commission and National Research Institute of Far Seas Fisheries. Seychelles: Victoria; 2007.
- [15] Kock KH, Purves MG, Duhamel G. Interactions between cetacean and fisheries in the Southern Ocean. *Polar Biology* 2006;29(5):379–88.
- [16] Hernandez-Milian G, Goetz S. Results of a short study of interactions of cetaceans and longline fisheries in Atlantic waters: environmental correlates of catches and depredation events. *Hydrobiologia* 2008;612:251–68.
- [17] Northridge S, Hofman R. Marine mammal interactions with fisheries. In: Reeves JR, editor. *Conservation and Management of Marine Mammals*. Washington: Smithsonian Institution Press; 1999. p. 99–119.
- [18] Hernandez-Milian G, Goetz S. Results of a short study of interactions of cetaceans and longline fisheries in Atlantic waters: environmental correlates of catches and depredation events. *Essential Fish Habitat Mapping in the Mediterranean* 2008:251–68.
- [19] Roche C, et al. Marine mammals and demersal longline fishery interactions in Crozet and Kerguelen Exclusive Economic Zones: an assessment of depredation levels. *CCAMLR Science* 2007;67–8214 2007:67–82.
- [20] Matkin C. Status of Prince William sound killer whales and the sablefish fishery in late 1987. Homer, AK: University of Alaska Sea Grant Marine Advisory Program; 1988.
- [21] Allen B, Angliss R. Sperm Whale (*Physeter macrocephalus*): North Pacific Stock. Alaska Marine Mammal Stock Assessments. NOAA TM-AFSC-206; 2008.
- [22] Angliss R, Outlaw R. Killer Whale (*Orcinus orca*): Eastern North Pacific Alaska Resident Stock. Alaska Marine Mammal Stock Assessments. NOAA TM-AFSC-168; 2010.
- [23] Anderson P. Marine mammals in the next one hundred years: Twilight for a Pleistocene megafauna. *Journal of Mammalogy* 2001;82(3):623–9.
- [24] Fearnbach H. Individual-based population assessment for cetaceans: using photographs to infer abundance, demography and individual quality, in Institute of Biological & Environmental Sciences. Scotland: University of Aberdeen; 2012.
- [25] Mesnick S, et al. Sperm whale population structure in the eastern and central North Pacific inferred by the use of single nucleotide polymorphisms, microsatellites and mitochondrial DNA. *Molecular Ecology Resources* 2011;11:278–98.
- [26] Thode A, et al. Sperm whale and longline fisheries interactions in the Gulf of Alaska—Passive acoustic component. North Pacific Research Board Final Report F09018.
- [27] Carothers C. A survey of US halibut IFQ holders: market participation, attitudes, and impacts. *Marine Policy* 2013;38:515–22.
- [28] Knapp G. Alaskan halibut captians' attitudes towards IFQs. *Marine Resource Economics* 1996;11:43–55.
- [29] Herrmann M, Criddle K. An econometric market model for the Pacific halibut fishery. *Marine Resource Economics* 2006;21(2):129–58.
- [30] Huppert DD, Best B. Study of supply effects on sablefish market price. School of Marine Affairs, University of Washington; 2004.
- [31] NMFS RAM Division, Pacific Halibut—Sablefish IFQ Report, 2012, National Marine Fisheries Service, Alaska Region Restricted Access Management, Juneau, AK. p. 1–97.
- [32] NOAA, Fisheries of the Exclusive Economic Zone off Alaska; North Pacific Halibut and Sablefish Individual Fishing Quota Cost Recovery Programs, Department of Commerce; 2011. p. 77209–13.
- [33] Williams GH. Incidental catch and mortality of Pacific halibut, 1962–2010. International Pacific Halibut Commission; 2011.
- [34] Hanselman D, Lunsford C, Rodgveller C. Assessment of the Sablefish Stock in Alaska. In: NMFS, editor. Juneau, AK: Alaska Fisheries Science Center; 2011.
- [35] Federal Registrar, Registered Buyer Standard Ex-Vessel Prices by Landing Location for 2011 IFQ Season. Department of Commerce, National Oceanic and Atmospheric Administration; 2011. p. 77209–14.
- [36] RAM Division, Individual Fishing Quota (IFQ) Allocations and Landings. Juneau, AK: National Marine Fisheries Service; 2012.
- [37] Ianelli J, Wilderbuer T, Nichol D. Assessment of Greenland Turbot in the Eastern Bering Sea and Aleutian Islands. Seattle, WA: National Marine Fisheries Service; 2011.
- [38] Jacobsen RB, Wilson DCK, Ramirez-Monsalve P. Empowerment and regulation—dilemmas in participatory fisheries science. *Fish and Fisheries* 2011:1–12.

- [39] Johannes RE, Freeman M, Hamilton RJ. Ignore fishers' knowledge and miss the boat. *Fish and Fisheries* 2000;1(3):257–71.
- [40] Nenadovic M, Johnson T, Wilson J. Implementing the Western Gulf of marine area closure: the role and perception of fishers' ecological knowledge. *Ecology and Society* 2012;17(1):20.
- [41] Bundy A, Davis A. Knowing in context: an exploration of the interface of marine harvesters' local ecological knowledge with ecosystem approaches to management. *Marine Policy* 2012;38:277–86.
- [42] Scholz A, et al. Participatory socioeconomic analysis: drawing on fishermen's knowledge for marine protected area planning in California. *Marine Policy* 2004;28(4):335–49.
- [43] Gilman E, Brothers N, Kobayashi DR. Principles and approaches to abate seabird by-catch in longline fisheries. *Fish and Fisheries* 2005;6(1):35–49.
- [44] Maynou F, et al. Estimating trends of population decline in long-lived marine species in the mediterranean sea based on fishers' perceptions. *PloS One* 2011;6(7):21818.
- [45] Beaudreau AH, Levin PS, Norman KC. Using folk taxonomies to understand stakeholder perceptions for species conservation. *Conservation Letters* 2011;4(6):451–63.
- [46] Yano K, Dahlheim M. Killer whale, *Orcinus-Orca*, depredation on longline catches of bottomfish in the southeastern Bering Sea and adjacent waters. *Fishery Bulletin* 1995;93(2):355–72.
- [47] Bernard RH. Research methods in anthropology. Qualitative and quantitative approaches. fourth ed. Lanham, MD: AltaMira Press; 2006.
- [48] Huntington H. Observations on the utility of the semi-directive interview for documenting traditional ecological knowledge. *Arctic* 1998;51(9):237–42.
- [49] Heeringa SG, West BT, Berglund PA. Applied survey data analysis. Taylor and Francis; 2010.
- [50] Strauss A, Corbin J. Grounded theory methodology. *Handbook of qualitative research*. 273–85.
- [51] Strauss A, Corbin J. Basics of qualitative research: techniques and procedures for developing grounded theory. Thousand Oaks, CA: Sage Publications; 2007.
- [52] Mackinson S. Integrating local and scientific knowledge: an example in fisheries science. *Environmental Management* 2001;27(4):533–45.
- [53] NMFS Alaska Regional Office, Current quota share holder and QS units, National Marine Fisheries Service, Seattle, WA; 2012.
- [54] Boone Jr. HN, Boone DA. Analyzing Likert data. *Journal of Extension* 2012;50:2.
- [55] Clason DL, Dormody TJ. Analyzing data measured by individual Likert-type items. *Journal of Agricultural Education* 1994;35:4.
- [56] Norman G. Likert scales, levels of measurement and the laws of statistics. *Advances in Health Sciences Education* 2010;15(5):625–32.
- [57] Lam T, Green KE, Bordignon C. Effects of Item grouping and position of the don't know option on questionnaire response. *Field Methods* 2002;14(4):418.
- [58] Agresti A. 2nd ed. Categorical data analysis, vol. 359. Gainesville, FL: Wiley-John and Sons Incorporated; 2002.
- [59] Fox J. Applied regression analysis and generalized linear models. Thousand Oaks, CA: Sage Publications; 2008.
- [60] Faraway JJ. Linear models with R, vol. 63. Boca Raton, FL: Chapman & Hall/CRC Press; 2004.
- [61] Clark J, Roberts J, Mees C. Depredation of fish caught on tuna lonlines in the BIOT area, in Workshop on the depredation in the tuna longline fisheries in the Indian Ocean, Victoria, Seychelles: Indian Ocean Tuna Commission and National Research Institute of Far Seas Fisheries; 2007.
- [62] Hucke-Gaete R, Moreno CA, Arata J. Operational interactions between cetaceans the the Patagonian toothfish (*Dissostichus eleginoides*) industrial fishery off southern Chile. *CCAMLR Science* 2004;11:127–40.
- [63] Tixier P, et al. Interactions of Patagonian toothfish fisheries with killer whales and sperm whales in the Crozet Islands Exclusive Economic Zone: an assessment of depredation levels and insights on possible mitigation strategies. *CCAMLR Science* 2010;17:179–95.